

# RODOLPHE ABOU ASSALI

**Nationality:** Lebanese

**Born on** August 4, 2000

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Personal Website

[R<sup>G</sup>](#), [ArXiv](#), [HAL](#)

## Education

- **2023–Present:** Joint PhD in Differential Geometry (3rd year)  
**Topic:** The biharmonic Steklov problem for differential forms  
**Supervisors:** [Nicolas Ginoux](#), [Georges Habib](#), [Samuel Tapie](#)  
Lebanese University, Faculty of Science, Lebanon – University of Lorraine, [Institut Élie Cartan](#), France.
- **2022–2023:** Master 2 in Fundamental and Applied Mathematics (Double degree)  
Lebanese University, Faculty of Science, Lebanon – University of Lorraine, [Institut Élie Cartan](#), France.
- **2021–2022:** Master 1 in Mathematics  
Lebanese University, Faculty of Science, Section II, Lebanon.
- **2018–2022:** University Diploma in Education  
Lebanese University, Faculty of Science, Section II, Lebanon.
- **2018–2021:** Bachelor's Degree in Mathematics  
Lebanese University, Faculty of Science, Section II, Fanar, Lebanon.

## Professional Experience

- **2025–2026:** Lecturer at IUT Nancy – Charlemagne – Computer Science Department  
**Courses taught:**
  - Descriptive Statistics (L1)
  - Numerical methods (L1)
  - Graph theory (L1)
  - Optimization Methods (L2, DACS)
- **2024–2025:** Lecturer at IUT Nancy – Charlemagne – Computer Science Department  
**Courses taught:**
  - Descriptive Statistics (L1)
  - Optimization Methods (L2, DACS)
  - Mathematical Complements (L3, RA–IL)

- **2022–2023:** Master 2 Internship in Differential Geometry  
**Topic:** Spectral Theory of the Biharmonic Steklov Operator  
**Supervisors:** [Nicolas Ginoux](#), [Georges Habib](#), [Samuel Tapie](#)  
 Lebanese University, Faculty of Science, Lebanon – University of Lorraine, [Institut Élie Cartan](#), France.
- **2018–2022:** Private tutoring – Mathematics and Physics for middle and high school students.

## Excellence Scholarships and Research Grants

- **2023–2026:** PhD scholarship from the [SAFAR](#) program of the French Embassy in Lebanon, for a PhD under the supervision of Nicolas Ginoux, Georges Habib, Samuel Tapie.
- **2025:** [DrEAM](#) Grant from the University of Lorraine under the Lorraine University of Excellence (LUE) initiative, for a two-month research stay at Sapienza University under the supervision of [Alessandro Savo](#).
- **2023:** Master Grant from the [Orion](#) program at the University of Lorraine for a 4-month research stay at the [IECL](#) under the supervision of [Nicolas Ginoux](#), [Georges Habib](#), [Samuel Tapie](#).

## Preprints

- **2026:** *Geometric eigenvalue estimates of Kuttler-Sigillito type on differential forms.* [arXiv:2602.09876 \[math.DG\]](#).
- **2025:** *Biharmonic Steklov problems with Neumann boundary conditions and spectral inequalities on differential forms.* [arXiv:2507.05049 \[math.DG\]](#).

## Talks

### Speaker

- **2025:** Seminar of the department of Basic and Applied Sciences for Engineering at Sapienza University, Rome  
*Biharmonic Steklov problems and spectral inequalities for differential forms.*
- **2025:** [Advanced PDE Seminar](#), Department of Mathematics and Statistics, University of Konstanz, Germany  
*Biharmonic Steklov problems and spectral inequalities for differential forms.*
- **2025:** [Differential Geometry Seminar](#), IECL Nancy  
*Biharmonic Steklov problems and spectral inequalities on differential forms.*
- **2024:** [IECL PhD Students Day](#), Nancy  
*The Biharmonic Steklov Operator.*
- **2024:** [Doctoral Forum of the EDST](#), Lebanese University, Beirut  
*The Biharmonic Steklov Problem on Differential Forms.*

## Attendee

- **2025:** Differential Geometry at L'Aquila ([DG@AQ25](#)).
- **2024:** European Doctorate School of Differential Geometry ([ESDG](#)), Granada.
- **2024:** Working group on the spectrum of the Steklov operator, IECL Nancy.
- **2023:** Geometry Day, IECL Nancy.

## Skills

### Computer Skills:

- **Programming languages:** C, CSS, C++, HTML, MATLAB (Beginner),  $\text{\LaTeX}$ .
- **Software:** Proteus 8 Professional, Tinkercad, Cisco Packet Tracer, Oracle Data Modeler (Intermediate).
- Adobe Photoshop, Illustrator, Dreamweaver, Visual Studio, VMware Workstation (Beginner).

### Language Skills:

- Arabic (Native language)
- French (Advanced level)
- English (Advanced level)

## Activities

- **2025–Present:** Member at Société Mathématique de France, ([SMF](#)).
- **2024–Present:** Co-organizer of the [PhD Students Seminar](#) at IECL.
- **2023–Present:** University teaching training sessions – Université de Lorraine.
- **2025:** Co-organizer of the [PhD days](#) with the PhD students at the University of Luxembourg.
- **2025:** Workplace First Aid Certificate (SST) – Nancy.
- **2024–2025:** [Math Club](#) at Lycée Jacques Callot, Vandœuvre-lès-Nancy.
- **2024–2025:** Math workshop at [the Science Festival](#) at FST and Jeanne d'Arc Elementary School.
- **2021–2023:** Student representative in M1 and M2.